

1. JK触发器

$$Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n$$

$$= J\bar{Q}^n + \bar{J}Q^n$$

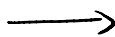
$$= J$$

$$= A \oplus B \oplus C \oplus D$$

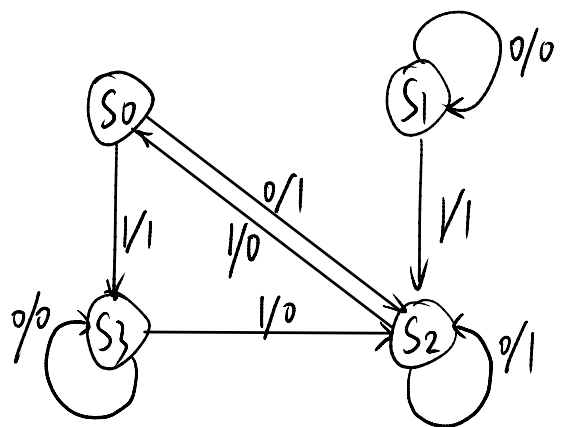
触发器置1的条件为A、B、C、D中高电平个数为奇数

2. 简化后的状态转移表:

PS	NS	
	X=0	X=1
S ₀	S ₂ , 1	S ₃ , 1
S ₁	S ₁ , 0	S ₂ , 1
S ₂	S ₂ , 1	S ₀ , 0
S ₃	S ₃ , 0	S ₂ , 0



状态转移图 (X/Z)



选择计数器法, 使用两个触发器, S₀、S₁、S₂、S₃ 分别编码为 00、01、10、11 (y₂ y₁)

X	y ₂	y ₁	y ₂ ⁿ⁺¹	y ₁ ⁿ⁺¹	Z
0	0	0	1	0	1
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	1	1	0
1	0	0	1	1	1
1	0	1	1	0	1
1	1	0	0	0	0
1	1	1	1	0	0

使用的是JK触发器,令 $K=J$

$$y_2^{n+1} = J_2 \bar{y}_2 + \bar{K}_2 y_2 = J_2 \bar{y}_2 + J_2 y_2 = J_2$$

$$= \bar{x} \bar{y}_2 \bar{y}_1 + \bar{x} y_2 \bar{y}_1 + \bar{x} y_2 y_1 + x \bar{y}_2 \bar{y}_1 + x \bar{y}_2 y_1 + x y_2 y_1$$

$$= \bar{x} (y_2 + \bar{y}_1) + x (\bar{y}_2 + y_1)$$

$$y_1^{n+1} = J_1 \bar{y}_1 + \bar{K}_1 y_1 = J_1$$

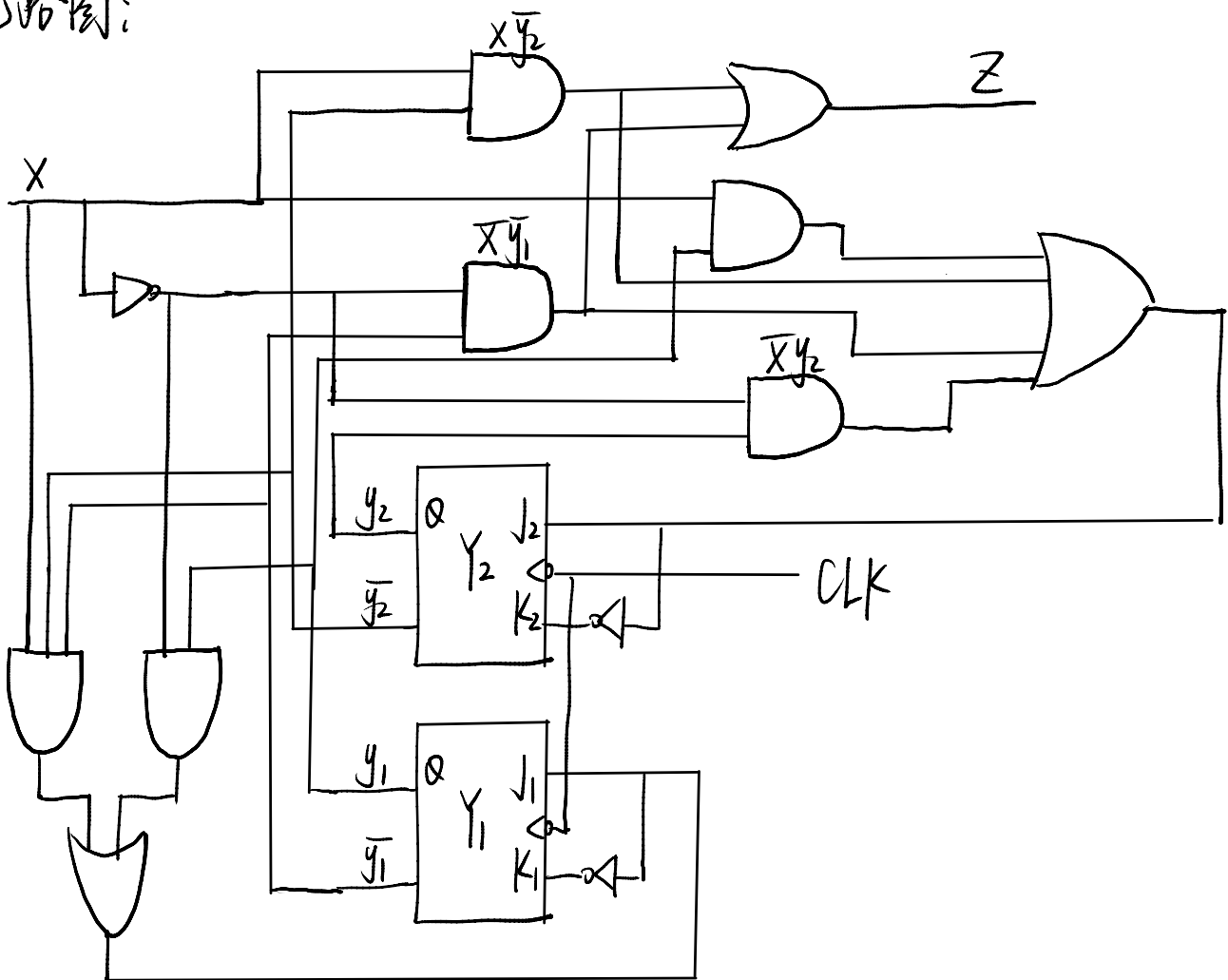
$$= \bar{x} \bar{y}_2 y_1 + \bar{x} y_2 y_1 + x \bar{y}_2 \bar{y}_1$$

$$= \bar{x} y_1 + x \bar{y}_2 \bar{y}_1$$

$$Z = \bar{x} \bar{y}_2 \bar{y}_1 + \bar{x} y_2 \bar{y}_1 + x \bar{y}_2 y_1 + x y_2 y_1$$

$$= \bar{x} \bar{y}_1 + x \bar{y}_2$$

电路图:



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}, module fsm(
    input clock,
    input k
);
    reg [1:0] state;
    parameter s0 = 2'b00,s1 = 2'b01,s2 = 2'b10,s3 = 2'b11;

    always @(posedge clock) begin
        case(state)
            s0: begin
                if(k)
                    ;
                else
                    state <= s1;
            end
            s1: begin
                if(k)
                    state <= s2;
                else
                    ;
            end
            s2:begin
                if(k)
                    ;
                else
                    state <= s3;
            end
            s3:begin
                if(k)
                    state <= s0;
                else
                    ;
            end
        endcase
    end
endmodule

```

4. 状态编码:

S₀: 主干道绿灯. 乡间道红灯: 00

S₁: 主干道黄灯. 乡间道红灯: 01

S₂: 主干道红灯. 乡间道绿灯: 10

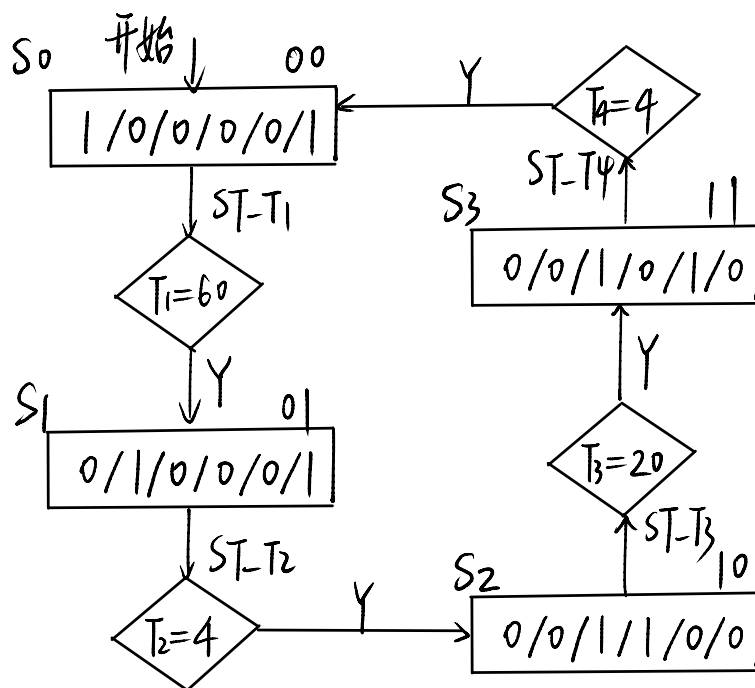
S₃: 主干道红灯. 乡间道黄灯: 11

输出信号共6个: LED1-G. LED1-Y. LED1-R. LED2-G. LED2-Y. LED2-R
分别表示主干道. 乡间道的绿. 黄. 红信号

采用两个D触发器表示状态 (Y₂ Y₁), 4个计时器 (T₁. T₂. T₃. T₄)

PS			NS			转移条件
状态名	Y ₂	Y ₁	状态名	Y ₂ (D)	Y ₁ (D)	
S ₀	0	0	S ₁	0	1	T ₁ =60
S ₁	0	1	S ₂	1	0	T ₂ =4
S ₂	1	0	S ₃	1	1	T ₃ =20
S ₃	1	1	S ₀	0	0	T ₄ =4

ASM流程图: (LED1-G/LED1-Y/LED1-R/LED2-G/LED2-Y/LED2-R)



次态方程:

$$Y_2(D) = \overline{Y_2} Y_1 (T_2=4) + Y_2 \overline{Y_1} (T_3=20)$$

$$Y_1(D) = \overline{Y_2} \overline{Y_1} (T_1=60) + Y_2 \overline{Y_1} (T_3=20)$$

计时器启动:

$$ST-T_1 = \overline{Y_2} \overline{Y_1}$$

$$ST-T_2 = \overline{Y_2} Y_1$$

$$ST-T_3 = Y_2 \overline{Y_1}$$

$$ST-T_4 = Y_2 Y_1$$

控制命令:

$$LED1-G = \overline{Y_2} \overline{Y_1}$$

$$LED1-Y = \overline{Y_2} Y_1$$

$$LED1-R = Y_2 \overline{Y_1} + Y_2 Y_1$$

$$LED2-G = Y_2 \overline{Y_1}$$

$$LED2-Y = Y_2 Y_1$$

$$LED2-R = \overline{Y_2} \overline{Y_1} + \overline{Y_2} Y_1$$

控制器

电路图:

